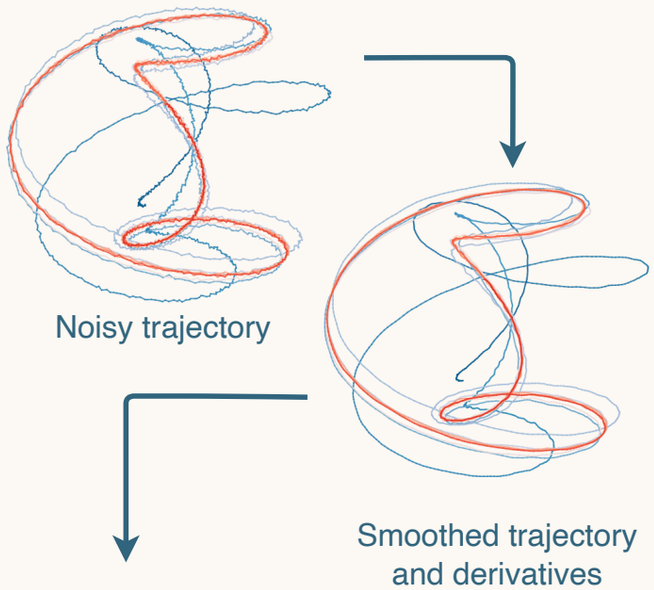
a Data Preparation

Smoothing and differentiation

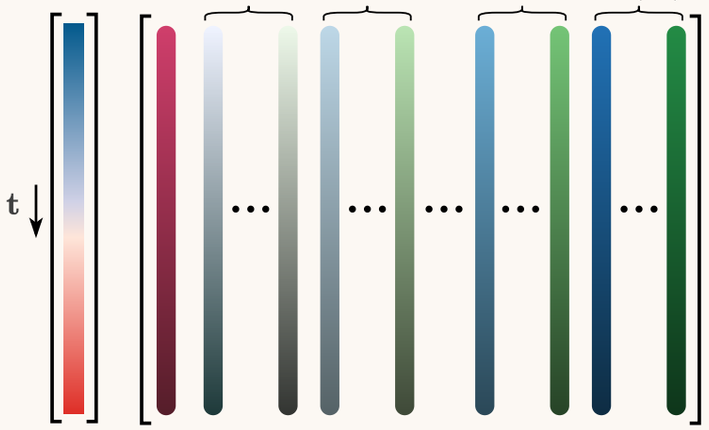
Adaptive Savitzky-Golay filter



$\mathbf{X}, \dot{\mathbf{X}}$

Design Matrix

$\Theta^{(0)}(\mathbf{X})$



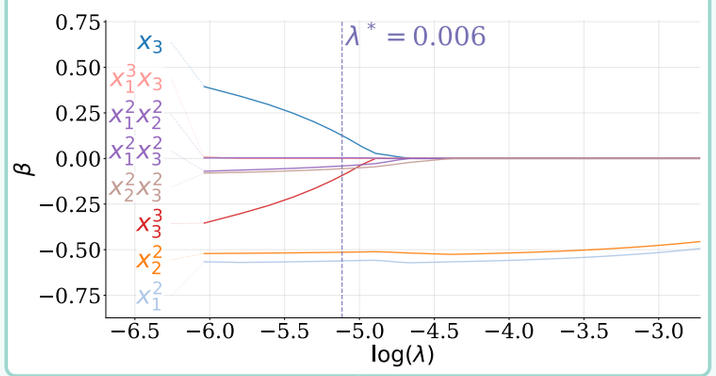
$\Theta^{(0)}(\mathbf{X})$

$$\dot{\mathbf{X}} = \Theta^{0}(\mathbf{X})\mathbf{B} + \mathbf{E}$$

b Frequentist Screening

Pass 1

Adaptive LASSO with ridge derived weights



Build OLS candidate models, and select best model with BIC

Design Matrix Refinement

Obtain $\Theta^{(1)}(\mathbf{X})$ including polynomials up to the maximum degree in the selected model

$\Theta^{(1)}(\mathbf{X})$

Pass 2

Adaptive LASSO with OLS derived weights

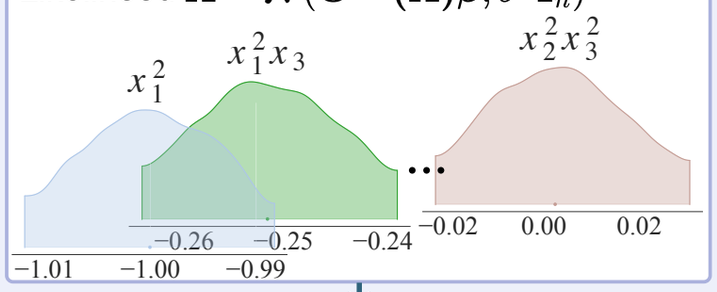
Build OLS candidate models, and select best model with BIC

$\Theta^{(2)}(\mathbf{X})$

c Bayesian Inference

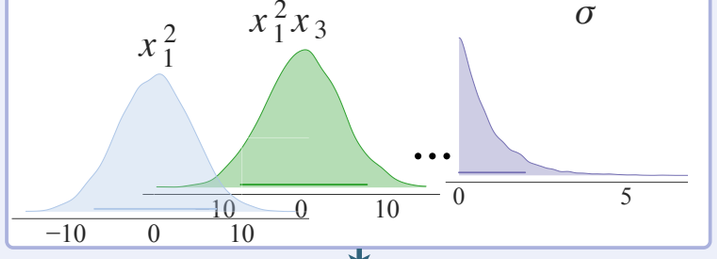
Bayesian Regression

Likelihood $\mathbf{X} \sim \mathcal{N}(\Theta^{(2)}(\mathbf{X})\beta, \sigma^2\mathbf{I}_n)$

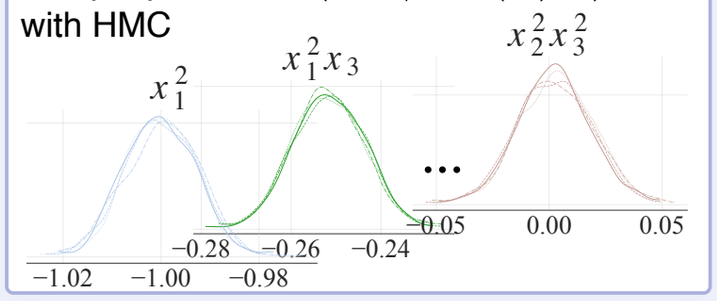


Priors

$\beta_k \sim \mathcal{N}(0, (2.5 s_y / s_{x_k})^2)$ $\sigma \sim Exp(1/s_y)$

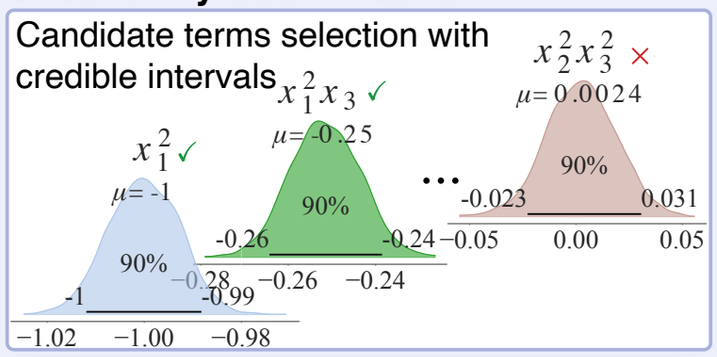


Sample posterior $p(\beta, \sigma \mid \Theta^{(2)}(\mathbf{X}), \mathbf{y})$ with HMC



Uncertainty Quantification

Candidate terms selection with credible intervals



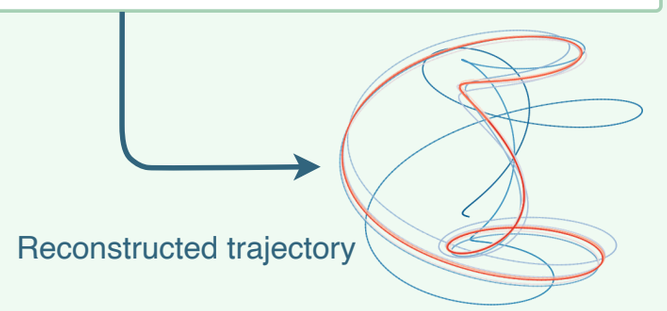
Equations

d Identified ODEs

$$\dot{x}_1 = -0.70x_1 + 1.00x_1x_3 - 3.50x_2$$

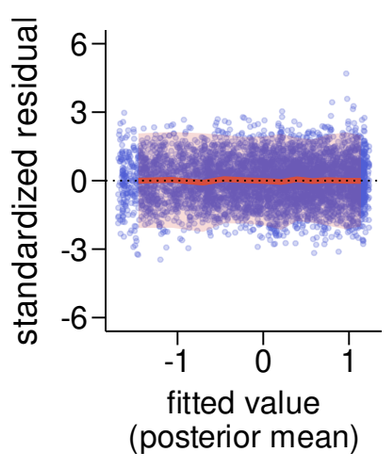
$$\dot{x}_2 = 3.50x_1 - 0.70x_2 + 1.00x_2x_3$$

$$\dot{x}_3 = 0.65 - 1.00x_1^2 - 0.25x_1^2x_3 + 0.10x_1^3x_3 - 1.00x_2^2 - 0.26x_2^2x_3 + 0.95x_3 - 0.33x_3^3$$

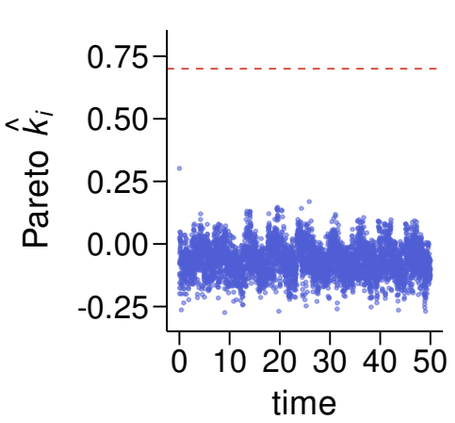


e Diagnostic Analysis

Residual analysis



Influential observations



Multicollinearity

